DIFFICULTY-STRATIFIED EVALUATION

Confidence Calibration in Large Language Models

Do language models know when they might be wrong, and does that get worse as questions get harder?

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PROBLEM STATEMENT

The Problem with Confident Answers

LLMs sound sure of themselves

Language models are used every day as study aids and decision support tools. Their answers sound fluent and confident even when the information behind them is wrong.

Calibration is the safeguard

Calibration asks a simple question: when a model says it is 95% confident, is it actually right about 95% of the time? A poorly calibrated model misleads the people who rely on it.

Difficulty may make it worse

Research on human judgment shows that overconfidence grows on difficult tasks (Moore & Healy, 2008). If LLMs behave the same way, their confidence is least reliable on the hardest questions.

Why it matters: people often overestimate how accurate LLMs are (Steyvers et al., 2025), so miscommunicated confidence carries real costs.

RESEARCH QUESTION & OBJECTIVE

How does task difficulty influence the calibration of verbally reported confidence in modern API language models, and does miscalibration grow as questions become harder?

Objective

Evaluate Claude Haiku, GPT-4o-mini, and Gemini 3.1 Flash-Lite on a difficulty stratified MMLU bank under one locked answer plus confidence protocol. Calibration is measured with accuracy, overconfidence gap, ECE, bootstrap confidence intervals, and Holm corrected difficulty regressions, with verbal and internal token confidence compared where available.

SOURCE LITERATURE REVIEW

Prior Research and the Gap

Human judgment

Moore & Healy (2008) found that people grow overconfident on hard tasks. Kruger & Dunning (1999) showed that limited skill often comes with limited insight into it.

ML calibration

Guo et al. (2017) showed that modern neural networks are often miscalibrated and introduced ECE. Desai & Durrett (2020) found that Transformers calibrate poorly out of domain.

LLM self-report

Kadavath et al. (2022), Lin et al. (2022), and Xiong et al. (2023) showed that LLMs can state confidence in words that partially tracks accuracy. Jiang et al. (2021) found QA models are often overconfident when wrong.

Knowing limits

Yin et al. (2023) found LLMs are worse than humans at spotting unanswerable questions. Steyvers et al. (2025) showed people overestimate LLM accuracy. Kuhn and Farquhar (2023, 2024) proposed semantic entropy as an alternative signal.

Gap: little student accessible work has tested how verbal confidence changes as difficulty rises across a standard benchmark, under one locked protocol applied to several API models.

METHOD

How the Study Was Run

Question bank

540 MMLU multiple choice items (180 frozen plus a 360 item supplement) covering STEM, Social Sciences, and Humanities, with easy, medium, and hard tiers fixed before evaluation.

Matched inference

3 API models, 5 trials per question, temperature 0.3, max 1,024 tokens, for 8,100 total responses. Every call used one locked prompt asking for ANSWER (A to D) and CONFIDENCE (0 to 100). With format repair retries, 100% of responses parsed.

Metrics

Accuracy, overconfidence gap (confidence minus accuracy), ECE with 10 bins, and bootstrap 95% confidence intervals from 2,000 resamples.

Statistics

Response level regression of the gap on difficulty, with six predeclared tests corrected using the Holm Bonferroni method. Answer token log probabilities were also collected for GPT-4o-mini.

MODELS EVALUATED

Claude Haiku

Anthropic API
claude-haiku-4-5

GPT-4o-mini

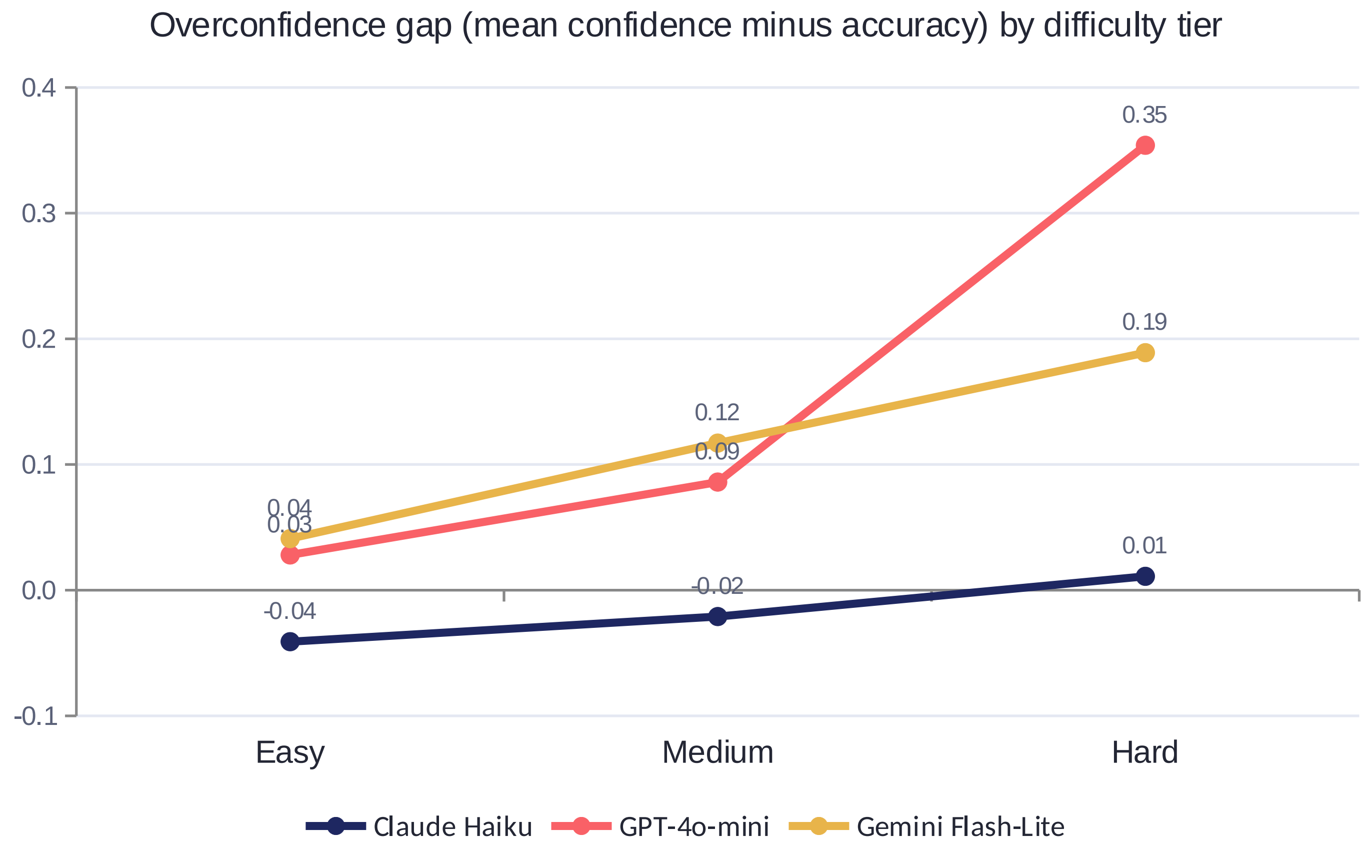
OpenAI API
gpt-4o-mini

Gemini 3.1 Flash-Lite

Google Gemini API
gemini-3.1-flash-lite

RESULTS & FINDINGS

Overconfidence Grows with Difficulty in All Three Models



A positive gap means overconfident and a negative gap means underconfident. Each point averages 900 responses.

0.354

GPT-4o-mini's gap on hard items. Accuracy falls to 51.4% while stated confidence stays near 87.

95.6%

of Gemini's confidence scores are 95 or higher, a pattern the paper calls confidence collapse

+0.011

Claude Haiku's gap on hard items. It stays close to calibrated at every difficulty level.

RESULTS & FINDINGS

Each Model Behaves Differently

Claude Haiku

Best calibrated

Accuracy	89.5%
ECE	0.033
Gap	−0.017

Slightly underconfident overall, with the lowest ECE in every domain (0.012 on STEM). Its stated confidence is comparatively informative.

GPT-4o-mini

Steepest miscalibration

Accuracy	71.5%
ECE	0.157
Gap	+0.156

Largest difficulty slope (0.163). Verbal confidence only weakly tracks internal token probability ($r = 0.22$), and internal ECE is actually worse (0.243).

Gemini Flash-Lite

Confidence collapse

Accuracy	87.6%
ECE	0.117
Gap	+0.115

Mean confidence of 99.2 with a standard deviation of 2.5. Stated confidence barely moves, so rising overconfidence comes almost entirely from falling accuracy.

All three difficulty effects stayed significant after Holm Bonferroni correction, and rankings held from the 180 item bank to the full 540 item bank.

LIMITATIONS

Limitations of This Study

Multiple-choice only

MMLU items only. Findings may not generalize to open ended generation or tool using agents.

Subject-level difficulty

Tiers come from a subject level mapping fixed in advance rather than item level IRT. One duplicate pair (q0060/q0073) was kept for archival comparability.

Verbal-first confidence

Internal token probabilities were available only for GPT-4o-mini. Claude and Gemini are excluded from that comparison because of API design, not data loss.

Constrained format

The locked ANSWER/CONFIDENCE format suppresses chain of thought and may lower absolute accuracy compared to CoT leaderboards. Claims concern calibration under this protocol.

No human baseline

No open weight local models and no formal human subjects arm were included in this submission.

Setting-specific

Results are specific to temperature 0.3, five trials, 1,024 max tokens, this 540 item bank, and July 2026 model versions.

CONCLUSIONS AND FURTHER UNDERSTANDING

Confidence Should Support Verification, Not Replace It

CONCLUSIONS

- All three models become more overconfident as difficulty rises. Accuracy drops on hard questions but stated confidence does not follow.
- Claude Haiku stays close to calibrated. GPT-4o-mini shows the steepest difficulty linked overconfidence. Gemini pairs high accuracy with confidence collapse.
- For GPT-4o-mini, verbal and internal confidence correlate only weakly, and neither one is a complete reliability measure.
- Calibration should be judged where models are most likely to fail, on hard items, not just where average performance looks strong.

FURTHER UNDERSTANDING AND NEXT STEPS

- Extend the study to harder benchmarks such as MMLU-Pro and GPQA.
- Add open weight local models like Llama and Mistral under the same locked protocol.
- Estimate difficulty at the item level and compare different ways of asking for confidence.
- Connect calibration metrics to how people actually rely on model answers.
- Repeat verbal versus internal comparisons wherever providers expose token probabilities.

AI USE TRANSPARENCY STATEMENT

How AI Tools Were Used in This Project

AI as the subject

Language model APIs from Anthropic, OpenAI, and Google were the subjects of the experiment. All inference calls, prompts, and raw outputs are archived in the project repository.

AI as an editing aid

An AI assistant was used only to help with LaTeX formatting and editing of the written brief.

Author-produced science

All experimental data, statistical analyses, figures, and tables came from the authors' own code (src/analyze_v2.py). Every claim and reference was verified by the authors against the original sources.